



SRI RAMAKRISHNA ENGINEERING COLLEGE

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Batch No:8

20EC279 – Mini Project I

Dept: ECE

AI ENHANCED ARRHYTHMIA DETECTION IN ECG DATA

Type of Project: Research /
Application

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Review

:Review II

Date

: 10.03.2025

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Introduction

- Arrhythmias can indicate serious heart issues, but they're not always easy to detect without expensive equipment or doctor's visit. Detecting them early can save lives.
- We use an PPG Sensor (which pick the values using IR light) connected to a microcontroller ESP32. This setup will collect heart data use machine learning model to spot the unusual heart rhythms.
- **Process:** First, we collect and pre process the BPM data, then train an AI model to recognize normal vs. abnormal heart rhythms.
- Then the patient is treated after detecting abnormal activities in their heart beats.

Problem identification, scope and objectives (Contd.,)

Problem Definition:

The aim of this project is to develop a portable, AI-powered device that uses ECG data to detect arrhythmias in real time. By embedding a trained AI model on a microcontroller, the device will continuously monitor heart activity for irregularities.

Scope

Embedded machine learning

Problem identification, scope and objectives

Objectives :

- To build a real-time arrhythmia detection model using Edge Impulse, an ESP32 microcontroller with ECG datasets from physionet.
- The project will involve training an AI model on ECG data in Edge Impulse, deploying the model to the EPS32 for on-device inference, and validating its ability to detect abnormal heart rhythms.
- This system will provide a compact and efficient solution for continuous heart monitoring.

Literature Survey - 01

Experiment	Inference	Problem gap
An improved hybrid AI model for prediction of arrhythmia using ECG signal	It combines deep learning and machine learning techniques, specifically using a two-layered Bidirectional Long Short-Term Memory (BiLSTM) network for feature extraction and a Random Forest algorithm for classification.	The paper lacks real-time deployment feasibility on edge devices and does not address model interpretability for clinical decision-making.

- **Source:**P. VaralakshmiP. Atshaya“An improved hybrid AI model for prediction of arrhythmia using ECG signal”Biomedical Signal Processing and Control.vol, issue October 2022

Literature Survey - 02

Experiment	Inference	Problem gap
A Hybrid Real-Time Automated System for the Detection of Congestive Heart Failure	The study observed that using shorter ECG data segments (2 seconds and 5 seconds) resulted in higher accuracy for detecting heart conditions compared to longer segments.	The paper lacks consideration of edge computing for low-latency processing and does not address personalized model adaptation for varying patient conditions in real-time CHF detection.

Source: M. A. Alzubaidi, "A Hybrid Real-Time Automated System for the Detection of Congestive Heart Failure", IEEE ACCESS vol, issue october 2022

Literature Survey – 03

Experiment	Inference	Problem gap
Prediction of Heart Disease Using Machine Learning	The article reviews various studies on using Machine Learning and Deep Learning techniques to predict heart disease.	Limited Databases: The study only considered five major databases, which may have excluded relevant articles from other sources

Source: Alfredo Daza Vergaray, “Prediction of Heart Disease Using Machine Learning”, journal of System and Management Sciences, issue 2023

Literature Survey - 04

Experiment	Inference	Problem gap
Automatic identification of rhythm/morphology abnormalities in 12-lead ECGs	The study focuses on developing a method for automatically identifying abnormalities in heart rhythms using 12-lead ECG data.	The paper lacks consideration of domain adaptation for diverse patient populations and does not address real-time processing constraints for large-scale ECG datasets.

- **Source** A. Kumar ,S. Choudhury“Automatic identification of rhythm/morphology abnormalities in 12- lead ECGs”, MethodsX vol, issue 2023

Literature Survey - 05

Experiment	Inference	Problem gap
A proposed technique for predicting heart disease using machine learning algorithms and an explainable AI method	The study also emphasizes the importance of feature correlation in improving predictions and mentions the development of a smartphone app for quick heart disease diagnosis	The paper lacks integration of real-time patient monitoring for dynamic risk assessment and does not explore the limitations of explainable AI in handling complex medical data.

- **Source:** Hosam El-Sofany Belgacem Bouallegue Yasser “A proposed technique for predicting heart disease using machine learning algorithms and an explainable AI method”, IEEE ACCESS vol, issue January2024

Existing technology of the project

- ❑ **ECG Signal Acquisition:** ECG signals are collected from the MIT-BIH Arrhythmia Dataset, which contains heart signals recorded from 47 individuals over 24 hours
 - ❑ **Preprocessing:** The ECG signals undergo denoising using Discrete Wavelet Transform (DWT) to remove unwanted noise and segmentation to extract individual heartbeats
 - ❑ **Feature Extraction:** Deep learning models like BiLSTM (Bidirectional Long Short-Term Memory) are used to automatically extract important features from the ECG signals
 - ❑ **Feature Reduction:** Techniques like Principal Component Analysis (PCA) are applied to reduce the number of features, making the model faster and more efficient
 - ❑ **Classification:** Random Forest, a machine learning algorithm, is used to classify the ECG signals into different arrhythmia types with high accuracy (up to 98.84%)
-
- **Source:** P. Varalakshmi and Atshaya P. Sankaran “An Improved Hybrid AI Model for Prediction of Arrhythmia Using ECG Signals”, IEEE ACCESS, issue 13 October 2022

Photographs

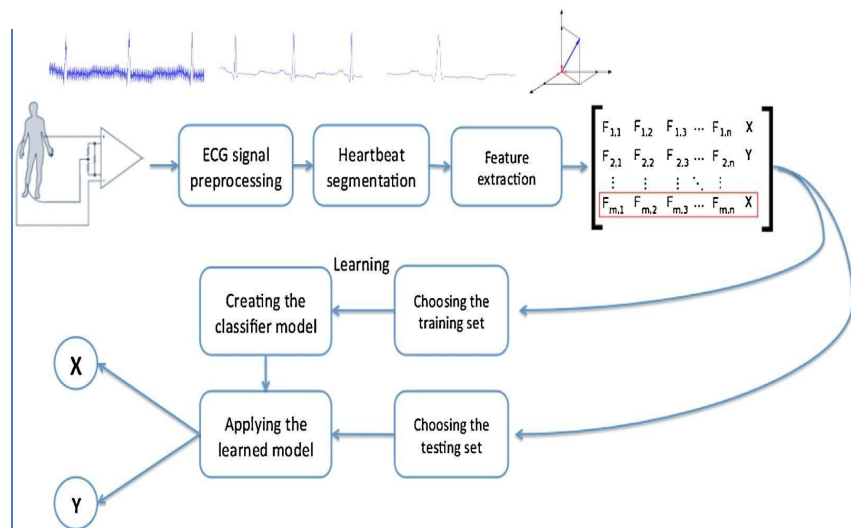
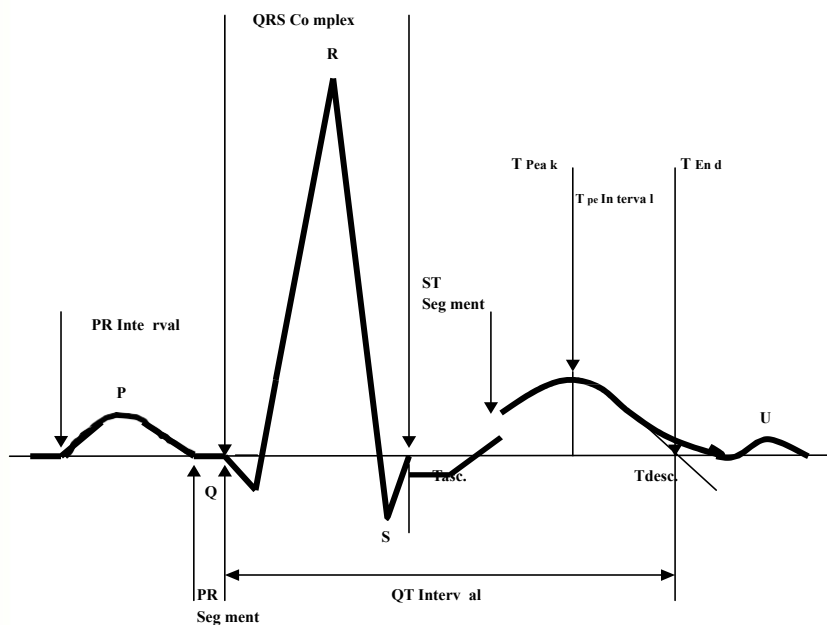
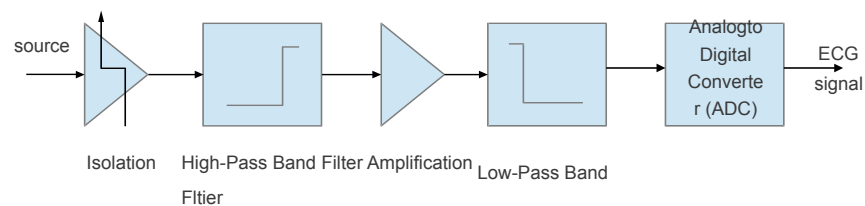


Fig.1 – A diagram of the arrhythmia classification system.



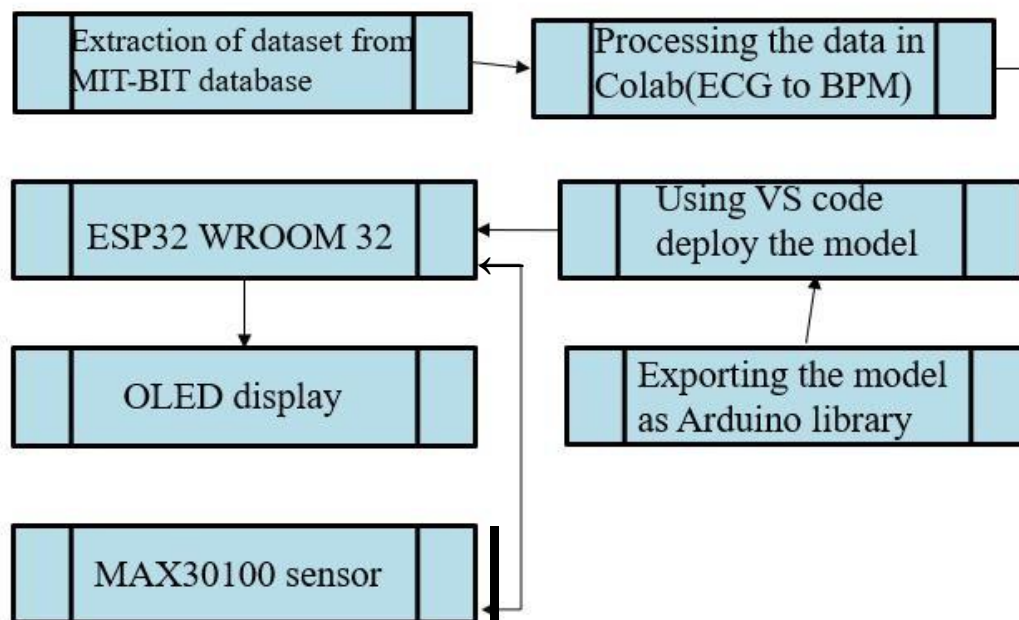
Methodology to be adopted

Flow Chart

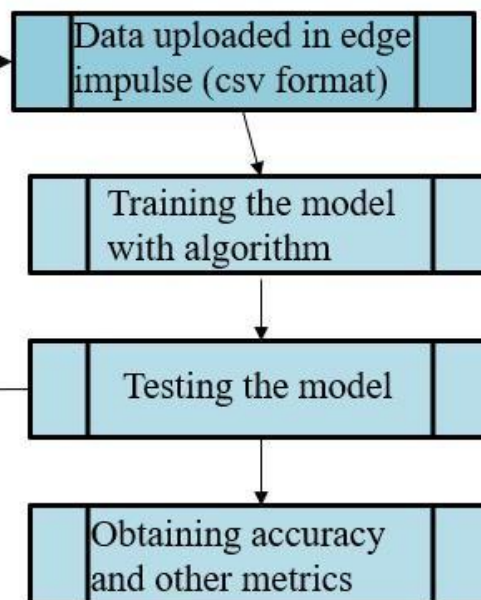


Design & Experimentation

HARDWARE SETUP



EDGE IMPULSE



METHODOLOGY

1. **Data Extraction:** Acquire ECG signals from the MIT-BIH database.
2. **Preprocessing in Colab:** Convert ECG data to BPM and filter noise.
3. **Edge Impulse Upload:** Import the CSV data for training/testing.
4. **Model Training:** Use chosen algorithms (classification etc.) to learn BPM patterns.
5. **Model Validation:** Evaluate accuracy, precision, recall, and other metrics.
6. **Arduino Library Export:** Convert the trained model into a microcontroller-ready library.
7. **ESP32 WROOM 32 Setup:** Integrate the library using PlatformIO (VS Code).
8. **MAX30100 Sensor:** Continuously capture PPG signals and convert them to BPM.
9. **OLED Display:** Show real-time BPM and classification results (e.g., Normal, Tachycardia).
10. **Live Inference:** Run the model on-device, verify accuracy via serial logs and display.

Results obtained till date

RESULT

Model
Model version: ? Quantized (int8)

Last training performance (validation set)

ACCURACY
98.4%

LOSS
0.04

Confusion matrix (validation set)

	BRADYCARDIA	NORMAL	TACHYCARDIA
BRADYCARDIA	100%	0%	0%
NORMAL	2.1%	97.9%	0%
TACHYCARDIA	0%	2.8%	97.2%
F1 SCORE	0.99	0.98	0.99

```

--- Classification Results ---
Bradycardia: 0.39%
Normal: 99.61%
Tachycardia: 0.00%
Predicted Class: Normal
Model Accuracy: 99.61%
System reset.

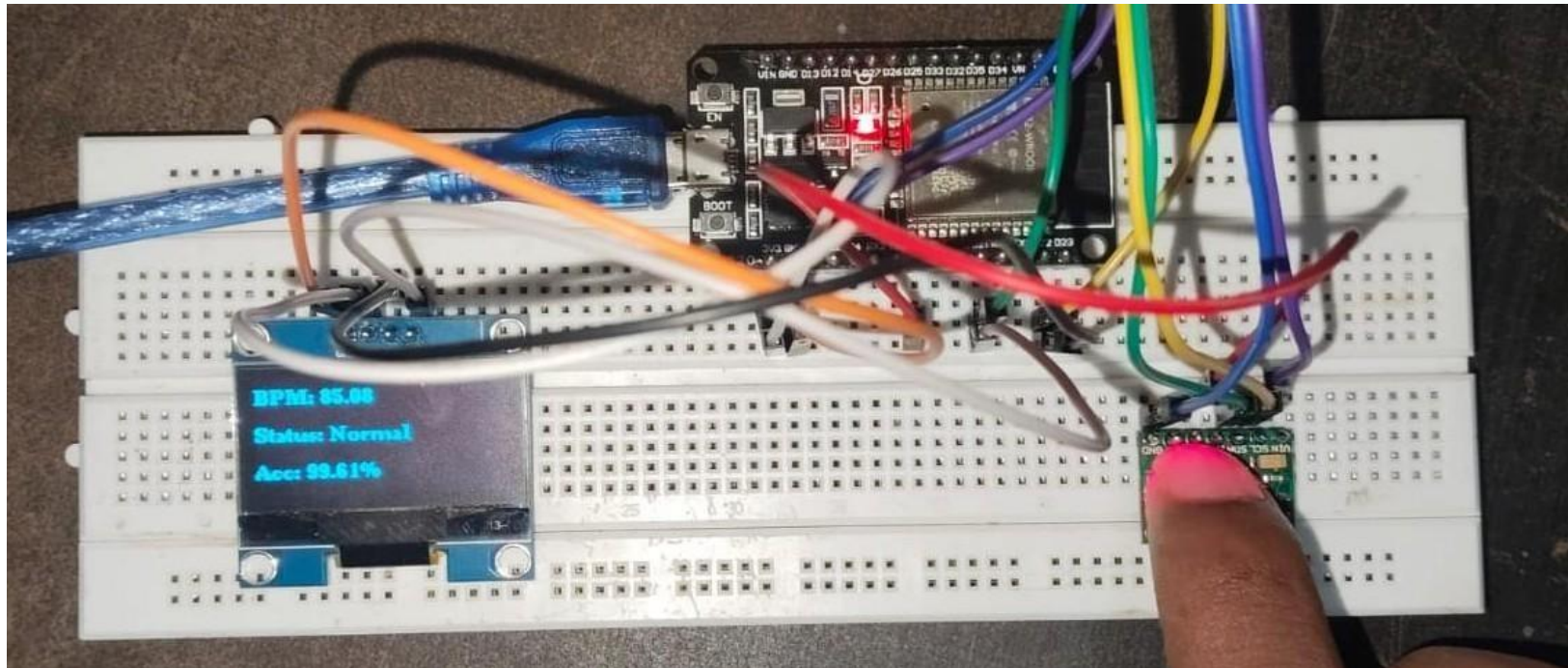
--- Classification Results ---
Bradycardia: 50.00%
Normal: 50.00%
Tachycardia: 0.00%
Predicted Class: Bradycardia
Model Accuracy: 50.00%

--- Classification Results ---
Bradycardia: 0.00%
Normal: 0.00%
Tachycardia: 99.61%
Predicted Class: Tachycardia
Model Accuracy: 99.61%

```

Accuarcy of software and hardware is compared and the result is obtained

Results obtained till date



A simple hardware setup for integrating the model into a embedded system.

Works to be carried out

WORKS TO BE CARRIED OUT

- PAPER WRITING IN PROCESS
- DOCUMENTATION AND REPORT WRITING

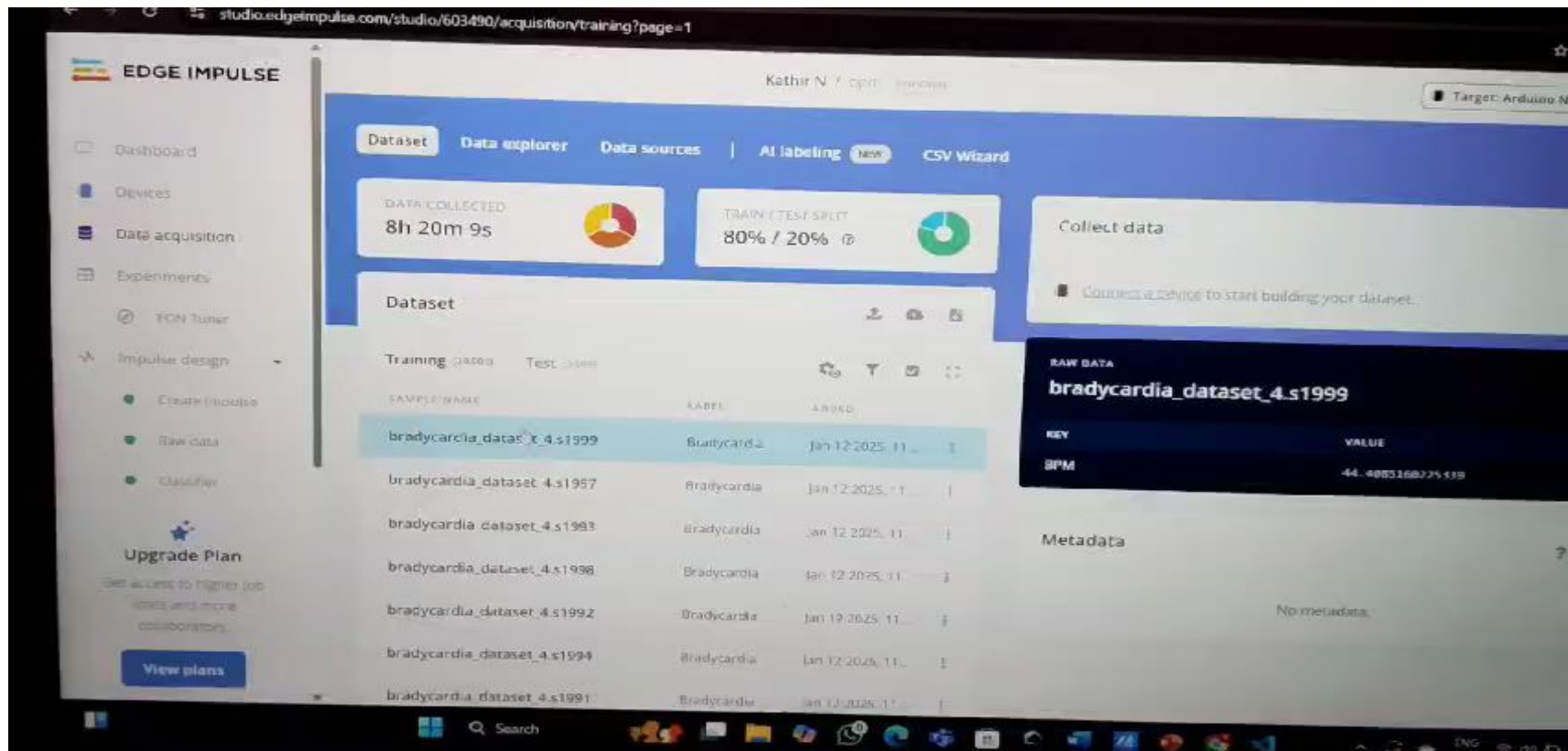
Timeline of the Project and Budget

Quarter Project Activity Timesheet

Chart Template with Months and Weeks



Proposed Project Outcome



A real-time heart rate monitoring system with high classification accuracy using ESP32 and MAX30100, powered by an embedded machine learning model from Edge Impulse.

Outcomes (Patent, Paper, Hackatons etc.,) – applied / published / granted

OUTCOME	STAGE	PERCENTAGE
PROJECT	YET TO BE COMPLETE D	SOFTWARE:100 % HARDWARE:90 %
PAPER	YET TO BE COMPLETE D	80% COMPLETED
HACKATHON	REGISTER FOR SHRISHTI	NOT SELECTED
	REGISTER FOR THE CREATATHON	SELECTED

1. P. Varalakshmi P. Atshaya "An improved hybrid AI model for prediction of arrhythmia using ECG signal" Biomedical Signal Processing and Control. vol, issue October 2022.
2. M. A. Alzubaidi, "A Hybrid Real-Time Automated System for the Detection of Congestive Heart Failure", IEEE ACCESS vol, issue october 2022.
3. Alfredo Daza Vergaray, "Prediction of Heart Disease Using Machine Learning", journal of System and Management Sciences, issue 2023
4. Kumar ,S. Choudhury "Automatic identification of rhythm/morphology abnormalities in 12- lead ECGs", MethodsX vol, issue 2023.
5. Hosam El-Sofany Belgacem Bouallegue Yasser "A proposed technique for predicting heart disease using machine learning algorithms and an explainable AI method", IEEE ACCESS vol, issue January 2024.